

On Three-Phase Reactive Power

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Abstract

It is well known that in single-phase alternating current (AC) electric power systems, both average power and reactive power, which are the real and imaginary parts of single-phase complex power, can be made to appear explicitly in the time-domain formulation of sinusoidal instantaneous power. However, unlike the single-phase case, in three-phase electric power systems, only (constant) three-phase average power appears in the three-phase instantaneous power counterpart. The mathematical reasons for this are explained and an algebraic reformulation of three-phase instantaneous power is contrived to unify complex power and instantaneous power across both cases.

I. Single-Phase Steady-State Power

Consider the single-phase steady-state sinusoidal inductive electric circuit of Figure 1.

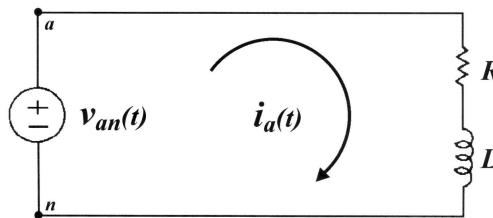


Figure 1. Single-phase steady-state sinusoidal inductive electric circuit

Let the ideal sinusoidal voltage source be:

$$v_{an}(t) = V_{peak} \cos(\omega t) \text{ V} \quad (1)$$

where: V_{peak} is the peak value of the sinusoidal voltage (in volts), ω is the radian frequency of the voltage in radians per second, t is time in seconds and the units of resistance R and inductance L are in ohms and henries, respectively.

The current is:¹

$$i_a(t) = I_{peak} \cos(\omega t - \theta) \text{ A} \quad (2)$$

where: $I_{peak} = V_{peak} / \sqrt{R^2 + (\omega L)^2}$ is the peak value of the sinusoidal current (in amperes), and $\theta = \tan^{-1}(\omega L/R)$ (in radians)

The instantaneous power (in watts) produced by the ideal voltage source is:²

$$p_a(t) = v_{an}(t)i_a(t) \quad (3)$$

Equation 3 may be expressed in terms of effective, root-mean-square (rms)³ voltage ($V_{rms} = \sqrt{2} V_{peak}$) and current ($I_{rms} = \sqrt{2} I_{peak}$) as:⁴

$$p_a(t) = V_{rms}I_{rms}\cos(\theta)[1 + \cos(2\omega t)] + V_{rms}I_{rms}\sin(\theta)[\sin(2\omega t)] \quad (4)$$

or:

$$p_a(t) = P_I(t) + Q_I(t) \quad (5)$$

To facilitate interpreting the meaning of Equations 4 and 5, its terms/factors are broken down as follows:

$$P_I(t) = P_I\alpha_I(t) \quad (6)$$

$$\text{where: } P_I = V_{rms}I_{rms}\cos(\theta) \quad (\text{constant}) \quad (7)$$

$$\text{and: } \alpha_I(t) = 1 + \cos(2\omega t) \quad (8)$$

$$Q_I(t) = Q_I\beta_I(t) \quad (9)$$

$$\text{where: } Q_I = V_{rms}I_{rms}\sin(\theta) \quad (\text{constant}) \quad (10)$$

$$\text{and: } \beta_I(t) = \sin(2\omega t) \quad (11)$$

Note that $p_a(t)$, $P_I(t)$ and $Q_I(t)$ as well as $\alpha_I(t)$ and $\beta_I(t)$ all vary with time whereas both P_I and Q_I are constant.

Figures 2 and 3 are temporal graphs of $P_I(t)$ and $Q_I(t)$, respectively, and, along with Equations 4 through 11, they reveal the following:⁵ $P_I(t)$ is a pulsating component of the instantaneous power $p_a(t)$ which is never negative (since $\alpha_I(t)$ is sinusoidal and biased by unity), ranges between $0 \leq P_I(t) \leq 2P_I$ and has an *average*^{*} value of $P_I = V_{rms}I_{rms}\cos(\theta)$. $Q_I(t)$ is another pulsating component of the instantaneous power $p_a(t)$ that is equally positive and negative, ranges between $-Q_I \leq$

^{*} The average value of $p_a(t)$ is $P_I = \frac{1}{T} \int_T p_a(t) dt = V_{rms}I_{rms}\cos(\theta)$, where $p_a(t)$ is periodic with period $T = \pi/\omega$, and $\pi = 3.14159...$ ⁶

$Q_I(t) \leq Q_I$ and has zero *average* value (because $\beta_I(t)$ is purely sinusoidal) with a *peak* value of $Q_I = V_{rms}I_{rms}\sin(\theta)$.

Because the average value of $Q_I(t)$ (which is purely sinusoidal) is zero whereas the average value of $P_I(t)$ is P_I , it follows that the average value of the instantaneous power $p_a(t)$ is also P_I . Therefore, $P_I = V_{rms}I_{rms}\cos(\theta)$ is the so-called *average power* associated with the circuit and $P_I(t)$ is the time-rate-of-change of unidirectional energy flowing from the circuit's ideal source to the resistor (R) of its RL -impedance load and not returning. $P_I(t)$ is temporally sinusoidal but positively biased by the *average power* $P_I = V_{rms}I_{rms}\cos(\theta)$.

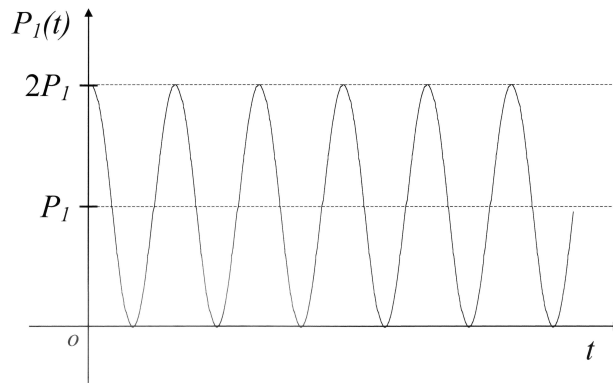


Figure 2. $P_I(t)$ waveform

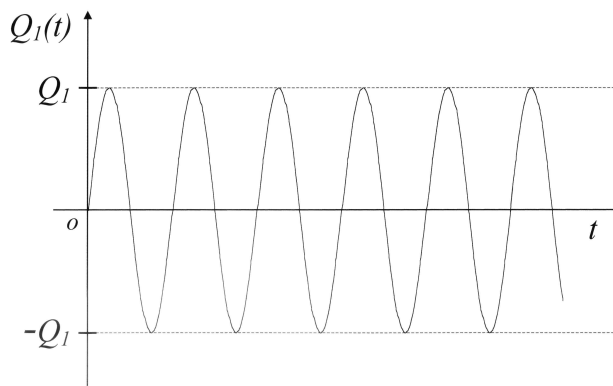


Figure 3. $Q_I(t)$ waveform

$Q_I(t)$ is the time-rate-of-change of bi-directional energy flowing (equally) back and forth between the circuit's ideal source and the inductor (L) of its RL -impedance load. The *peak* value of $Q_I(t)$ is $Q_I = V_{rms}I_{rms}\sin(\theta)$ whereas its *average* value is zero. $Q_I = V_{rms}I_{rms}\sin(\theta)$ is the so-called *reactive power* associated with the circuit.

Complex power captures both average power and reactive power (both of which are scalars) in the following complex *number*:⁷

$$S_I = P_I + jQ_I \quad (\text{Cartesian form}) \quad (12)$$

$$S_I = S_I e^{j\theta} \quad (\text{polar form}) \quad (13)$$

where: $j = \sqrt{-1}$ ^{*} and e is the transcendental number 2.71828...¹⁰

The scalar magnitude (S_I) of the (polar form) complex power is known as the *apparent power*¹¹ and is:[†]

$$S_I = |S_I| = |P_I + jQ_I| = V_{rms}I_{rms} \quad (14)$$

so that:

$$P_I = \text{Re}(S_I) = |S_I| \cos(\theta) = S_I \cos(\theta) = V_{rms}I_{rms}\cos(\theta) \quad (15)$$

$$\text{and: } Q_I = \text{Im}(S_I) = |S_I| \sin(\theta) = S_I \sin(\theta) = V_{rms}I_{rms}\sin(\theta) \quad (16)^\ddagger$$

Note that average power (P_I), reactive power (Q_I) and apparent power (S_I) are all scalars, each having units corresponding to the product of voltage and current (i.e., $V_{rms}I_{rms}$). But the unit of voltage, i.e., one volt, is one joule of energy per coulomb of electric charge¹⁷ and the unit of electric current, i.e., one ampere, is the charge flow-rate of one coulomb per second of time.¹⁸ So the product of these units yields a unit of one joule of energy per second, which is one watt of power. So instantaneous power, average and reactive power (and therefore complex power) as well as apparent power, all have units of joules per second (watts). To help prevent mistaking them for one another (i.e., to promote rather than obscure their differences), it is common practice to use the (artifact) units of watts for instantaneous and average power, VA (volt-amperes) for apparent and complex power, and VAR (volt-amperes reactive, also frequently appears as VAr in the literature) for reactive power. “To quote an early power engineer: ‘We didn’t know what watts were; volt-amperes were a measure of power. If anything happened so that the power did not get into the volt-amperes—well, that was the manufacturer’s fault. Nobody raised any questions, and nobody answered any.’ P.N. Nunn, *General Electric Review*, Sept. 1956, p. 43.”¹⁹

Recapitulating, both the average power P_I and the reactive power Q_I appear in the instantaneous power $p_a(t)$ as follows (from Equations 4, 6 and 9):

$$p_a(t) = P_I[1 + \cos(2\omega t)] + Q_I[\sin(2\omega t)] \quad (17)$$

However, as will be shown, this is not the case for three-phase instantaneous power.

^{*} The imaginary number⁸⁻⁹ $j = \sqrt{-1}$ more commonly appears as (letter) i in mathematics and physics literature.

[†] Because $P_I = V_{rms}I_{rms}\cos(\theta)$ and $Q_I = V_{rms}I_{rms}\sin(\theta)$ constitute the perpendicular sides of a right (power) triangle,¹² this result follows from the Pythagorean Theorem. Apparent power is, inter alia, useful as a means of rating the capability of certain AC electrical apparatus (e.g.; transformers and synchronous machines).¹³

[‡] Since $P_I = \text{Re}(S_I)$ and $Q_I = \text{Im}(S_I)$, they are also known as real and imaginary power, respectively.¹⁴ They are also known as *active*¹⁵ and *quadrature*¹⁶ power, respectively.

II. Three-Phase Steady-State Power

Consider the three-phase steady-state sinusoidal inductive electric circuit of Figure 4. The circuit consists of three single-phase circuits connected in the so-called grounded wye configuration. Each of these circuits is the same as the single-phase circuit of Figure 1 except for two of the voltage sources. The sources, which are said to form a balanced, positive-sequence²⁰ three-phase set, are all equal in magnitude, phase displaced from one another by $2\pi/3$ radians and sum to zero. These line-to-neutral (the neutral is grounded) voltages are:

$$v_{an}(t) = V_{peak} \cos(\omega t) \text{ V.} \quad (18)$$

$$v_{bn}(t) = V_{peak} \cos(\omega t - 2\pi/3) \text{ V.} \quad (19)$$

$$v_{cn}(t) = V_{peak} \cos(\omega t + 2\pi/3) \text{ V.} \quad (20)$$

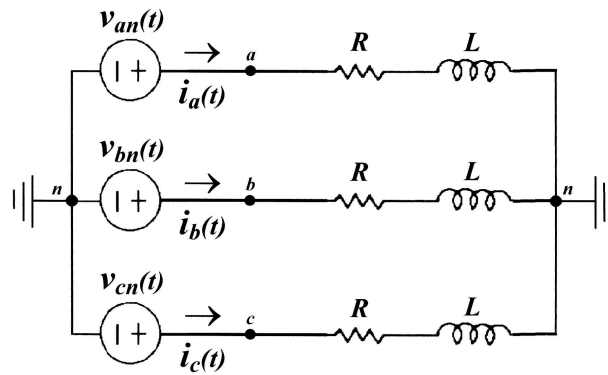


Figure 4. Three-phase steady-state sinusoidal inductive electric circuit

The corresponding line (phase) currents are also a balance positive-sequence set and are:

$$i_a(t) = I_{peak} \cos(\omega t - \theta) \text{ A.} \quad (21)$$

$$i_b(t) = I_{peak} \cos(\omega t - \theta - 2\pi/3) \text{ A.} \quad (22)$$

$$i_c(t) = I_{peak} \cos(\omega t - \theta + 2\pi/3) \text{ A.} \quad (23)$$

where the definition of θ can be found after Equation 2.

The three-phase instantaneous power is:

$$p_3(t) = p_a(t) + p_b(t) + p_c(t) \quad (24)$$

where $p_a(t)$ is given by Equation 3 and the other two instantaneous phase powers are:

$$p_b(t) = v_{bn}(t)i_b(t) \quad (25)$$

$$p_c(t) = v_{cn}(t)i_c(t) \quad (26)$$

The three-phase instantaneous power is known to be:²¹

$$p_3(t) = 3V_{rms}I_{rms}\cos(\theta) \quad (\text{constant}) \quad (27)$$

Because $p_3(t)$ is constant (and therefore equals its own average power, P_3), one may intuit that there is no three-phase reactive power phenomena (since none appears in Equation 27). However, this cannot be true because the overall electrical phenomena (physics) for each (electrically independent) phase of the three-phase circuit is the same as that for the single-phase circuit (Figure 1) except for timing (e.g.; the voltages and currents are out of phase with one another by $2\pi/3$ radians). It follows then that there *is* reactive power associated with each phase. In fact, the reactive power flowing in one of the three phases (i.e., phase *a*) is described by Equations 9, 10 and 11. Reactive power does not appear in Equation 27 because, as will be shown, the relative timing of the time-varying reactive powers flowing in the three phases is such that, like balanced three-phase voltages and currents, they form a balanced three-phase set which sums to zero. Thus, according to the late Professor Elgerd: "...the concept of 'three-phase reactive power' $Q_{3\phi}$ makes as little physical sense as would the concept of a 'three-phase current' ..." ²²

Substituting Equation 7 into Equation 27 yields:

$$P_3 = 3V_{rms}I_{rms}\cos(\theta) = 3P_1 \quad (28)$$

So three-phase average power (P_3) is three times single-phase power. *Assuming* ^{*} the same (three-to-one) relationship between three- and single-phase reactive powers yields:

$$Q_3 = 3Q_1 = 3V_{rms}I_{rms}\sin(\theta) \quad (29)$$

Ergo, the three-phase complex power counterpart artifact is:

$$S_3 = P_3 + jQ_3 = 3P_1 + j3Q_1 = 3(P_1 + jQ_1) = 3S_1 \quad (30)$$

From which it follows that three- and single-phase apparent powers are related by:

$$S_3 = |S_3| = |3S_1| = 3|S_1| = 3S_1 = 3V_{rms}I_{rms} \quad (31)$$

Thus the mathematical apparatus of single-phase complex power is extrapolated to three-phase complex power.

^{*} According to Professor Elgerd: "The only reason for so doing is to obtain formula symmetry between real and reactive powers." ²³

Three-phase complex power and its constituents P_3 and Q_3 can be expressed in terms of rms line-to-line voltage and rms line current.* The rms value of each of the line-to-line voltages $v_{ab}(t)$, $v_{bc}(t)$ and $v_{ca}(t)$ is (where V_{rms} is the rms line-to-neutral voltage):²⁴

$$V_{ll, rms} = \sqrt{3} V_{rms} \quad (32)$$

Because the circuit of Figure 4 is (grounded) wye connected, the rms value of each of the line current currents $i_a(t)$, $i_b(t)$ and $i_c(t)$ is:²⁵

$$I_{l, rms} = I_{rms} \quad (33)$$

Substituting $3 = \sqrt{3} \times \sqrt{3}$, $V_{rms} = V_{ll, rms} / \sqrt{3}$ (from Equation 32) and Equation 33 into Equation 31 yields:

$$S_3 = |S_3| = \sqrt{3} \times \sqrt{3} (V_{ll, rms} / \sqrt{3}) I_{l, rms} = \sqrt{3} V_{ll, rms} I_{l, rms} \quad (34)$$

From Equations 28, 29 and 30,

$$S_3 = P_3 + jQ_3 = 3V_{rms}I_{rms}\cos(\theta) + j3V_{rms}I_{rms}\sin(\theta) \quad (35)$$

Substituting Equation 34 into Equation 35 and dropping the rms subscript notation yields:

$$S_3 = P_3 + jQ_3 = \sqrt{3} V_{ll}I_l\cos(\theta) + j\sqrt{3} V_{ll}I_l\sin(\theta) \quad (36)$$

so that:

$$P_3 = \sqrt{3} V_{ll}I_l\cos(\theta) \quad (37)$$

$$Q_3 = \sqrt{3} V_{ll}I_l\sin(\theta) \quad (38)$$

$$S_3 = |S_3| = \sqrt{3} V_{ll}I_l \quad (39)$$

Having described the extrapolation of single-phase complex power formulae and its constituent parts to three-phase complex power and its constituent parts,[†] the interpretation of three-phase instantaneous power (Equation 27) is now reconsidered. The goal is to extrapolate single-phase instantaneous power to three-phase instantaneous power in such a way as to unify complex power and instantaneous power across both (single- and three-phase) cases. This is accomplished as follows.

* As a practical matter, the measurement of this voltage and current does not require access to a neutral and/or ground.

[†] Equations 36 through 39 are valid independent of connection (wye, grounded wye or delta).²⁶

First, consider the derivation of Equation 27 from Equation 24, where $p_a(t)$ is given by Equation 3 and the other two instantaneous phase powers, $p_b(t)$ and $p_c(t)$, are given by Equations 25 and 26. Substituting Equations 19 and 22 into Equation 25 yields:

$$p_b(t) = V_{peak} \cos(\omega t - 2\pi/3) I_{peak} \cos(\omega t - 2\pi/3 - \theta) \quad (40)$$

Equation 4, with ωt replaced by $\omega t - 2\pi/3$, yields:

$$P_b(t) = V_{rms} I_{rms} \cos(\theta) [1 + \cos(2\omega t - 4\pi/3)] + V_{rms} I_{rms} \sin(\theta) [\sin(2\omega t - 4\pi/3)] \quad (41)$$

But $\cos(2\omega t - 4\pi/3) = \cos(2\omega t + 2\pi/3)$ and $\sin(2\omega t - 4\pi/3) = \sin(2\omega t + 2\pi/3)$. Substitution of these trigonometric identities yields:

$$P_b(t) = V_{rms} I_{rms} \cos(\theta) [1 + \cos(2\omega t + 2\pi/3)] + V_{rms} I_{rms} \sin(\theta) [\sin(2\omega t + 2\pi/3)] \quad (42)$$

By similar reasoning:

$$P_c(t) = V_{rms} I_{rms} \cos(\theta) [1 + \cos(2\omega t - 2\pi/3)] + V_{rms} I_{rms} \sin(\theta) [\sin(2\omega t - 2\pi/3)] \quad (43)$$

Substituting Equations 4, 42 and 43 into Equation 24 yields:

$$p_3(t) = V_{rms} I_{rms} \cos(\theta) [3 + \cos(2\omega t) + \cos(2\omega t - 2\pi/3) + \cos(2\omega t + 2\pi/3)] + V_{rms} I_{rms} \sin(\theta) [\sin(2\omega t) + \sin(2\omega t - 2\pi/3) + \sin(2\omega t + 2\pi/3)] \quad (44)$$

but:

$$\cos(2\omega t) + \cos(2\omega t - 2\pi/3) + \cos(2\omega t + 2\pi/3) = 0 \quad (\text{balanced three-phase}) \quad (45)$$

likewise:

$$\sin(2\omega t) + \sin(2\omega t - 2\pi/3) + \sin(2\omega t + 2\pi/3) = 0 \quad (\text{balanced three-phase}) \quad (46)$$

Substituting these last two equations into Equation 44 yields Equation 27 (QED).

A scheme for extrapolating single-phase instantaneous power to three-phase instantaneous power in such a way as to (mathematically) unify complex power and instantaneous power across both (single- and three-phase) cases entails writing the last substitution result (i.e., Equation 27) as:

$$p_3(t) = 3V_{rms} I_{rms} \cos(\theta) [1] + V_{rms} I_{rms} \sin(\theta) [0] \quad (47)$$

Multiplying Equation 46 by three yields:

$$3[\sin(2\omega t) + \sin(2\omega t - 2\pi/3) + \sin(2\omega t + 2\pi/3)] = 0 \quad (48)$$

Replacing zero in the second bracketed term in Equation 47 with the left side of Equation 48 yields:

$$p_3(t) = 3V_{rms}I_{rms}\cos(\theta)[1] + 3V_{rms}I_{rms}\sin(\theta)[0] \quad (49)$$

or (by Equations 28 and 29):

$$p_3(t) = P_3[1] + Q_3[0] \quad (50)$$

Employing Equations 37 and 38, this result in terms of (rms) line-to-line voltage and (rms) line current is:

$$p_3(t) = \sqrt{3} V_{ll}I_l\cos(\theta)[1] + \sqrt{3} V_{ll}I_l\sin(\theta)[0] \quad (51)$$

Finally, by way of Equation 51 and the following seven equations, the unification is (by analogy to Equations 4 through 11):

$$p_3(t) = P_3(t) + Q_3(t) \quad (\text{constant}) \quad (52)$$

$$P_3(t) = P_3\alpha_3(t) \quad (\text{constant}) \quad (53)$$

$$\text{where: } P_3 = \sqrt{3} V_{ll}I_l\cos(\theta) \quad (\text{constant}) \quad (54)$$

$$\text{and: } \alpha_3(t) = 1 \quad (\text{constant}) \quad (55)$$

$$Q_3(t) = Q_3\beta_3(t) \quad (\text{constant}) \quad (56)$$

$$\text{where: } Q_3 = \sqrt{3} V_{ll}I_l\sin(\theta) \quad (\text{constant}) \quad (57)$$

$$\text{and: } \beta_3(t) = 0 \quad (\text{constant}) \quad (58)$$

Note that *all* of the expressions above are constants. The key to the unification provided herein is essentially the introduction (by way of Equation 58) of the $\beta_3(t)$ artifact as an algebraic artifice so as to prevent the three-phase reactive power Q_3 from being annihilated by $\beta_3(t)$'s zero equivalent counterpart factor (Equation 58). In other words, $\beta_3(t)$ is a contrivance introduced to symbolise zero²⁷ and thus prevent Q_3 from becoming nullified (i.e., rendered invisible) as a result of substituting Equation 46 into Equation 44 in obtaining Equation 27.

III. Conclusion

The relationship between single-phase instantaneous power and its complex-power (and complex power's constituent parts) has been reviewed. This has included showing how, in single-phase alternating current (AC) electric power systems, both average power and reactive power, which

are the real and imaginary parts of single-phase complex power, can be made to appear explicitly in the time-domain formulation of sinusoidal instantaneous power. In addition, a physics interpretation of the attendant mathematics has also been reviewed.

The well-known result regarding how in three-phase power systems, and unlike in single-phase systems, only (constant) three-phase average power appears in the three-phase instantaneous power counterpart (i.e., three-phase reactive power does not so appear) has also been reiterated; and the mathematical reasons for this have been reviewed. Finally, an algebraic reformulation of three-phase instantaneous power has been presented which unifies complex power and instantaneous power across both cases. While this reformulation is admittedly contrived, it is motivated by good intentions, namely, to foster the further understanding of electric power engineering.

IV. Epilogue

Those who are not specialists/practitioners in the field of electric power engineering might reasonably ask why one should continue to strive to better understand reactive power? There are (at least) two reasons. The first is the intellectual growth to be acquired through further study of perhaps one of the more mathematically arcane concepts of the field.

The second is practical. The world's electricity infrastructure is based on single- and three-phase AC (alternating current) electric power systems in which reactive power is inherent, ubiquitous and is of considerable import. Reactive power is intrinsic to nearly all apparatus found in the three traditional sectors (generation, transmission and distribution) of electric power systems and the importance of controlling reactive power is well known in the field.²⁸⁻²⁹ "The reactive power characteristics of various power-system components are summarized as follows: reactive power is generated by over-excited synchronous machines, capacitors, cables, lightly loaded overhead lines; and absorbed by under-excited synchronous machines, induction motors, inductors, transformers, heavily loaded overhead lines."³⁰ "There are more induction motors in regular use than any other type of electric motor."³¹ "Approximately 70 percent of all electrical energy generated in the United States is used to drive electric motors."³²

That reactive power has economic impacts can be demonstrated by considering the inductive circuit of Figure 1 as a simple^{*} prototypical electric power system, where the electricity supply (generation), transmission system and load (demand) are represented by the ideal voltage source, resistor and inductor, respectively. Even though the purely inductive load consumes no (net) energy (but is solely responsible for the circuit's reactive power), because current flows (Equation 2), there *is* average power (and therefore energy) consumed by the transmission system (i.e., by way of *Joule heating*³³ of the resistor). It follows that the supply system (ideal voltage source) produces this average power at a cost dependent on the ultimate source of the input energy (fuel) as-

* Simplifications include at least the following. By modelling the supply as an ideal voltage source, the internal (Thevenin) impedance (synchronous reactance) of the generator is neglected. Representing transmission by resistance alone neglects inductive reactance and (capacitive) line charging associated with transmission lines. The purely inductive load consumes no average power, which is a deliberate contrivance here for illustrative purposes, but is also impractical since this load consumes no (net) energy (and so it generates no revenue).

sociated with its energy conversion process. In addition to this economic consideration, there is a (generally undesirable) voltage drop across the transmission system that would not occur without the current drawn by the purely reactive load.

And so with all this in mind, it is the author's hope that the ideas and concepts compiled herein contribute to fostering the further understanding of reactive power.

V. Acknowledgement

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